

Chemical Reactions and Equations

What a chemical reaction is, how to write and balance a chemical equation, the five main types of reactions with NCERT examples, and why corrosion and rancidity matter in daily life.

Introduction

A chemical reaction happens when one or more substances change into new substances. The starting substances are called reactants. The new substances formed are called products. You can often see that a reaction has taken place by a change in colour, a change in temperature, evolution of a gas, or formation of a precipitate.

Examples from NCERT: magnesium ribbon burns in air to form magnesium oxide, zinc reacts with dilute sulphuric acid to give hydrogen gas, and lead nitrate solution forms a yellow precipitate with potassium iodide solution.

Chemical Equations and Balancing

A chemical equation uses symbols and formulae to show what reacts and what is formed. A word equation can be written first, then replaced by formulae.

Example: Zinc + Sulphuric acid gives Zinc sulphate + Hydrogen.



Balanced Chemical Equation

A balanced chemical equation has the same number of atoms of each element on both sides. This follows the Law of Conservation of Mass. Mass is neither created nor destroyed in a chemical reaction. So the total mass of reactants equals the total mass of products.

An unbalanced equation is called a skeletal equation. To balance it, change only the coefficients (the numbers in front of formulae). Never change the subscripts inside a formula.

Worked Example 1, Balancing

Balance: $\text{Fe} + \text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$

Step 1. Count atoms. Left: Fe = 1, H = 2, O = 1. Right: Fe = 3, O = 4, H = 2.

Step 2. Balance Fe and O first. Put 3 before Fe on the left and 4 before H₂O.



Now check: Left Fe = 3, O = 4, H = 8. Right Fe = 3, O = 4, H = 8. The equation is balanced.

NCERT also shows how to make an equation more useful. You can add symbols for physical states: (s) solid, (l) liquid, (g) gas, (aq) aqueous. You can show that a gas is evolved by (g) or an upward arrow, and that a precipitate is formed by (s) or a downward arrow. You can also write conditions above the arrow, such as heat or catalyst.

Types of Chemical Reactions

NCERT Class 10 groups reactions into five main types. Learn the definition and one example for each.

Exothermic and endothermic reactions

An exothermic reaction releases heat, as in $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{heat}$. An endothermic reaction absorbs energy; most decomposition reactions require heat, light or electricity.

1. Combination Reaction

Combination Reaction

Two or more reactants combine to form a single product.

Examples:

- Calcium oxide with water: $\text{CaO(s)} + \text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(aq)} + \text{heat}$. This is used for white washing. Calcium hydroxide reacts slowly with CO_2 in air to form calcium carbonate, which gives the shine on walls.
- Burning of coal: $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$
- Formation of water: $2\text{H}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{H}_2\text{O(l)}$

2. Decomposition Reaction

Decomposition Reaction

A single compound breaks down into two or more simpler products. Heat, light or electricity is needed.

- Thermal decomposition: $2\text{FeSO}_4\text{(s)} \rightarrow \text{Fe}_2\text{O}_3\text{(s)} + \text{SO}_2\text{(g)} + \text{SO}_3\text{(g)}$ on heating. The green colour of ferrous sulphate changes to brown ferric oxide.
- $\text{CaCO}_3\text{(s)} \rightarrow \text{CaO(s)} + \text{CO}_2\text{(g)}$ on heating. This is used in lime manufacture.
- Photolysis: $2\text{AgCl(s)} \rightarrow 2\text{Ag(s)} + \text{Cl}_2\text{(g)}$ in sunlight. White silver chloride turns grey. This reaction is used in black and white photography.
- Electrolysis of water: $2\text{H}_2\text{O(l)} \rightarrow 2\text{H}_2\text{(g)} + \text{O}_2\text{(g)}$ on passing electric current.

3. Displacement Reaction

Displacement Reaction

A more reactive element displaces a less reactive element from its compound.

- $\text{Fe(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{FeSO}_4\text{(aq)} + \text{Cu(s)}$. Iron displaces copper. Blue copper sulphate turns light green.
- $\text{Zn(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{ZnSO}_4\text{(aq)} + \text{Cu(s)}$

4. Double Displacement Reaction

Double Displacement Reaction

Two compounds exchange ions and form two new compounds. If one product is an insoluble solid, it is called a precipitate, and the reaction is called a precipitation reaction.

- $\text{Na}_2\text{SO}_4(\text{aq}) + \text{BaCl}_2(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{NaCl}(\text{aq})$. Barium sulphate is a white precipitate.
- $\text{Pb}(\text{NO}_3)_2(\text{aq}) + 2\text{KI}(\text{aq}) \rightarrow \text{PbI}_2(\text{s}) + 2\text{KNO}_3(\text{aq})$. Lead iodide is a yellow precipitate.

Worked Example 2, Identify the Type

Classify: (a) $2\text{Pb}(\text{NO}_3)_2(\text{s}) \rightarrow 2\text{PbO}(\text{s}) + 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ on heating, (b) $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$

(a) Lead nitrate breaks into several products on heating. So it is thermal decomposition.

(b) Hydrogen removes oxygen from copper oxide. Copper oxide is reduced and hydrogen is oxidised. So it is a redox reaction. It is also a displacement in terms of oxygen.

Oxidation and Reduction (Redox)

Redox Reaction

If a substance gains oxygen or loses hydrogen during a reaction, it is oxidised. If a substance loses oxygen or gains hydrogen, it is reduced. When oxidation and reduction happen together, the reaction is a redox reaction.

- $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$. Here CuO loses oxygen, so CuO is reduced to Cu. H₂ gains oxygen, so H₂ is oxidised to H₂O.
- $\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + 2\text{H}_2\text{O} + \text{Cl}_2$. MnO₂ is reduced, HCl is oxidised.

In terms of the modern view used at higher classes, oxidation is loss of electrons and reduction is gain of electrons. For Class 10, the oxygen and hydrogen definition is enough.

Worked Example 3, Redox

Identify the substance oxidised and reduced: $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$

ZnO loses oxygen to form Zn, so ZnO is reduced. C gains oxygen to form CO, so C is oxidised.

Corrosion and Rancidity

These are effects of oxidation in daily life.

Corrosion and Rancidity

Corrosion is the slow eating away of a metal when it reacts with air and moisture. Rancidity is the bad smell and taste that appears when oils and fats in food get oxidised.

- **Corrosion:** Rusting of iron. $4\text{Fe} + 3\text{O}_2 + x\text{H}_2\text{O} \rightarrow 2\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ (hydrated iron oxide). Silver gets a black coating of silver sulphide. Copper gets a green coating of basic copper carbonate. Corrosion damages bridges, ships and iron objects.
- **Prevention of corrosion:** Painting, oiling, greasing, galvanising (coating iron with zinc), chrome plating, anodising or making alloys.
- **Rancidity:** Fried food kept for a long time smells bad because fats are oxidised. Prevention: keep food in air tight containers, keep in a refrigerator, add anti-oxidants, flush food packets with nitrogen instead of air (as in chips packets).

Summary

- A chemical equation must be balanced so that atoms are conserved.
- Combination: $\text{A} + \text{B} \rightarrow \text{C}$. Example: $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$.
- Decomposition: $\text{AB} \rightarrow \text{A} + \text{B}$. Needs heat, light or electricity.
- Exothermic reactions release heat; endothermic reactions absorb energy.

- Displacement: $A + BC \rightarrow AC + B$. A more reactive element displaces a less reactive one.
- Double displacement: $AB + CD \rightarrow AD + CB$. Precipitate may form.
- Redox: oxidation (gain of oxygen or loss of hydrogen) and reduction (loss of oxygen or gain of hydrogen) occur together.
- Corrosion is oxidation of metals. Rancidity is oxidation of fats and oils. Both can be slowed by keeping air away.

Practice Questions

1. Balance the equation: $Al + CuCl_2 \rightarrow AlCl_3 + Cu$.

2. What is observed when ferrous sulphate crystals are heated in a test tube? Write the balanced equation.

3. Identify the type: $Na_2SO_4(aq) + BaCl_2(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$. Why is it called a precipitation reaction?

4. A brown coating is seen on iron articles kept in moist air. Name the process and write its chemical name. Give two methods to prevent it.

5. Give one example each of thermal decomposition and photolysis with balanced equations.

6. In the reaction $CuO + H_2 \rightarrow Cu + H_2O$, which substance is oxidised and which is reduced? Why?